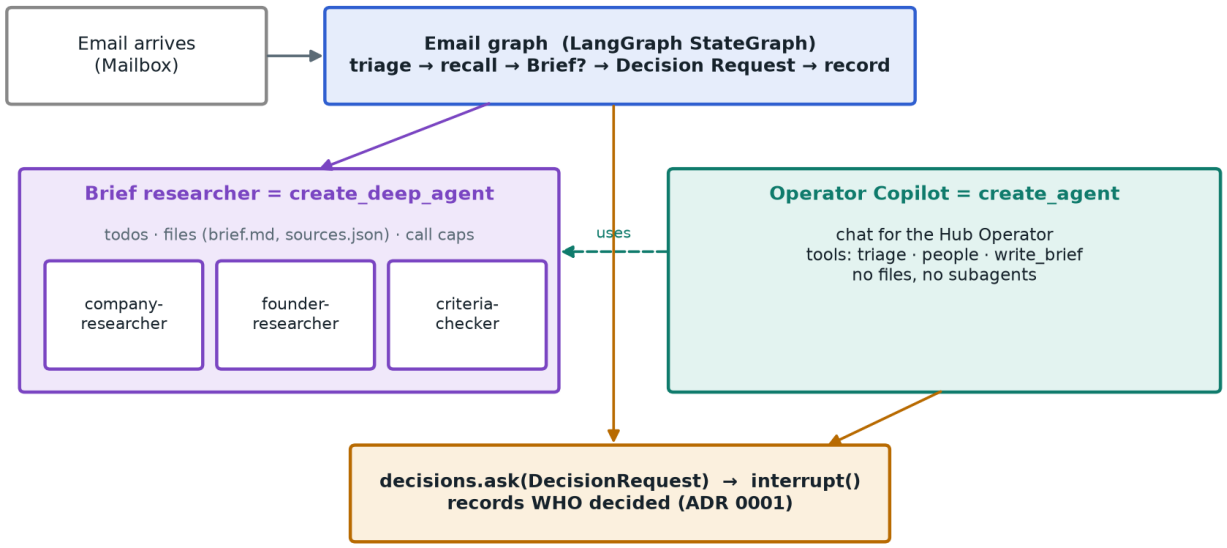
Deep Agents vs our own harness

AI Gothenburg Ops Copilot · research summary · 5 Oct 2026 · deepagents 0.7.20, langchain 1.4.3, langgraph 1.2.12. Full sourced report: docs/research/deep-agents-vs-own-harness.md

The answer in five lines

- **Brief researcher:** keep Deep Agents, but configured tightly (exact version pin, explicit models, call caps, restricted file tools).
- **Operator Copilot:** use plain `create_agent`. It needs no subagents or files.
- **Decisions:** go through our own `decisions.ask()`, not Deep Agents' built-in approval, so every Decision records its Hub Operator.
- **Budget:** one Brief is about 15–25 model calls, so Ling's free 100/day covers about 4–6 Briefs.
- **Biggest risk:** Agent Inbox, our planned demo screen, is archived and looks incompatible. We need another surface.

Recommended shape: Deep Agent only where research needs it



1. What Deep Agents actually is

`create_deep_agent()` is not a separate runtime. It stacks middleware on top of LangChain's `create_agent()` and returns a normal LangGraph graph, so it runs on LangGraph Server and traces to LangSmith with no glue.

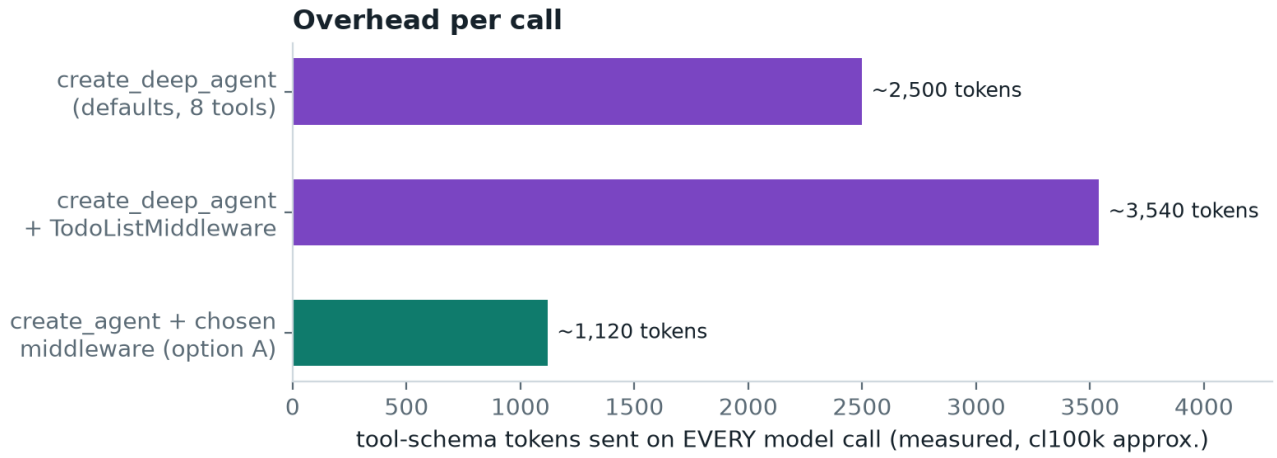
Feature	What it does	Default?
Subagents (task tool)	Each subagent sees only its task text and returns one report. Raw search results never reach the main context.	yes
Virtual filesystem	ls, read_file, write_file, edit_file, delete, glob, grep. Files live in graph state, so they are saved in checkpoints.	yes
Planning (write_todos)	A to-do list the agent keeps up to date.	no, removed in 0.7.0
Summarization	Shrinks long histories at 85% of the context window (170k tokens if the model has no profile).	yes
Human approval	interrupt_on = LangChain's HumanInTheLoopMiddleware.	opt-in
general-purpose subagent	An extra subagent with every tool.	yes, turn it off
Prompt caching (Anthropic)	Added always; does nothing on non-Claude models.	yes

2. Strengths and weaknesses

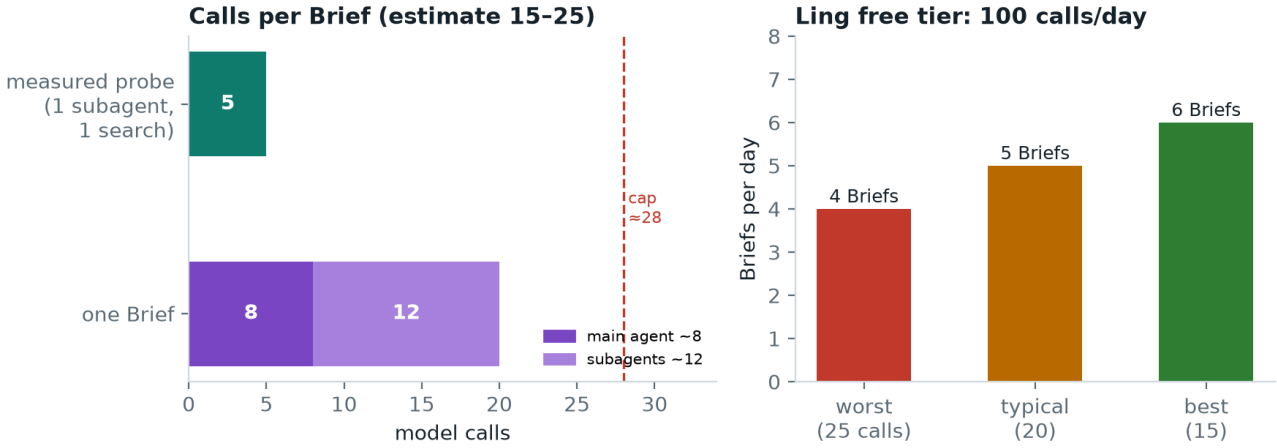
Good for us	Watch out
Context isolation for the 3 researchers, for free	Weekly releases with breaking changes (0.x)
Files match our artifacts: brief.md, sources.json, previous_brief.md	write_file overwrites and delete is on: it could wipe previous_brief.md
Researchers can run in parallel	No call limit by default (recursion limit 9,999)
Default prompt is now empty: our 'no score, cite everything' rules don't fight it	A model name string defaults to OpenAI's Responses API; Command Code only does chat completions
Works with scripted fake models in tests	Main-agent middleware does not reach subagents: caps must be set in each one
Plain LangGraph: Server, Studio, LangSmith just work	Middleware is invisible in Studio's graph view, only in traces

3. Cost per call and per Brief

Every model call resends the tool definitions. The full Deep Agents bundle costs about twice as much per call as picking only the pieces we need.

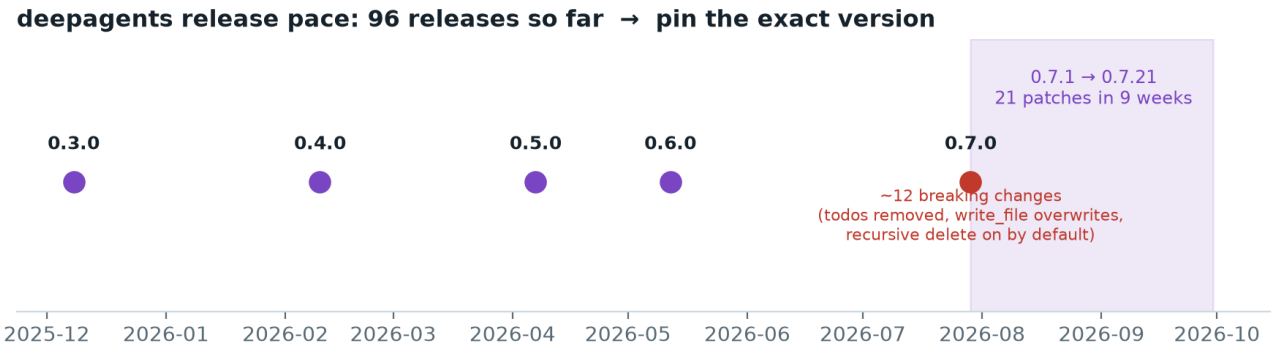


A measured run with 1 subagent and 1 search made 5 model calls. A real Brief with 3 researchers is estimated at 15–25. With caps of about 10 calls for the main agent and 6 per researcher, the hard ceiling is about 28.



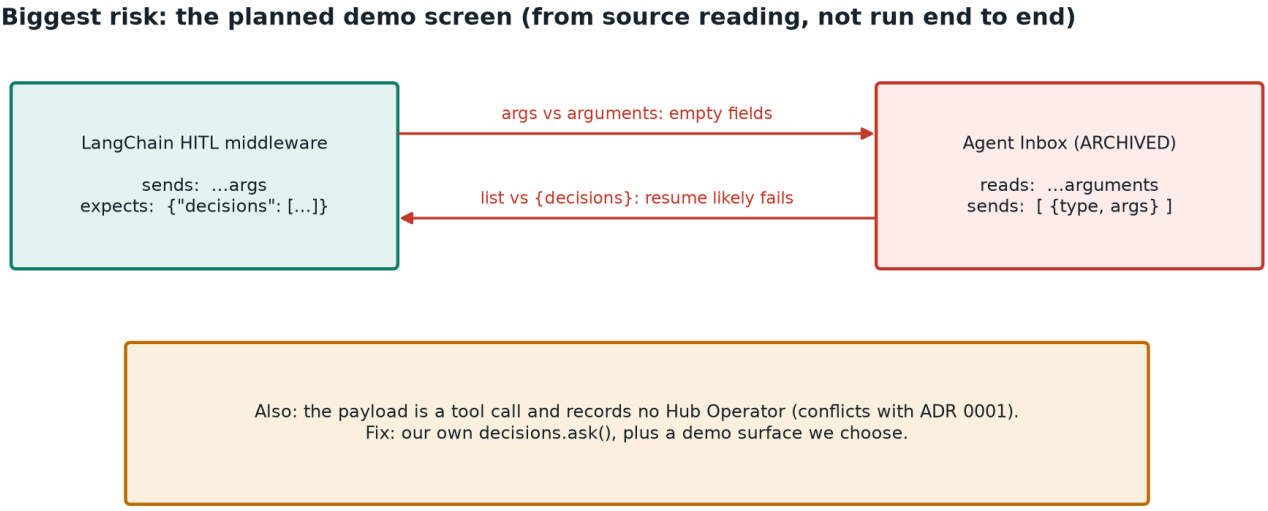
Weak models re-plan and retry more, so expect the high end with Ling. The release profile (Claude) has no daily request cap.

4. Release pace



Our `pyproject.toml` says `deepagents>=0.7.20`, which is a floor, not a pin. Recommendation: pin the exact version and keep Deep Agents behind `briefs.write_brief()`, so an upgrade only ever touches one module.

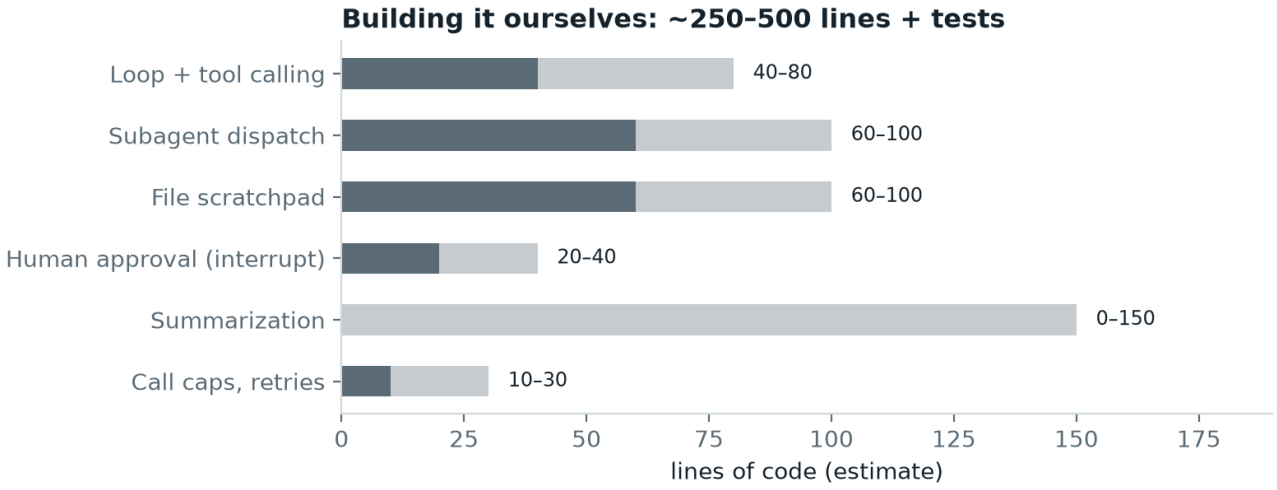
5. The biggest risk: our demo screen



Confirmed on GitHub: **langchain-ai/agent-inbox is archived**. LangChain's other UI, Agent Chat UI, is still active. The format mismatches come from reading the source, not from running it end to end.

6. Could we just build it ourselves?

waku-agent's loop is about 95 lines, but it is tied to the Anthropic SDK and has no persistence, approvals, subagents or summarization. Matching what the Brief researcher needs would take roughly 250–500 lines plus tests.



Option	Verdict
A. create_agent + chosen middleware	Works (measured). Half the tokens per call. You still import deepagents pieces, so you still feel its churn.
B. create_deep_agent, configured	**Recommended for Briefs.** Keeps the Deep Agents learning goal and its subagent defaults.
C. Our own LangGraph subgraph	Best once the Brief shape stops changing: fewest calls, most predictable. Too early for v1.
D. Deep Agents for Briefs, create_agent for the Copilot	**Recommended overall.**

7. What to check first when we build it

- A scripted Brief writes brief.md and sources.json, never changes previous_brief.md, and never calls delete.
- A looping researcher is stopped by its own call cap; a looping main agent is stopped too.
- Researchers only get ls, read_file, write_file and their search tool.
- Claude through Command Code accepts the prompt-caching blocks Deep Agents adds (1 real request).
- Ling's real context window, so summarization triggers before the provider errors.
- Decision Request round trip records the Hub Operator, and we pick the demo surface that replaces Agent Inbox.